DATA ANALYTICS

Laboratory Assignment

B.Tech – Electronics & Communication Engineering

Subject	Data Analytics (Theory + Lab)
Semester	As Applicable
Total Questions	6 (Attempt All)
Tools	Google Colab · Tableau · Power BI · AWS · Intelligent Cloud
Datasets	HR, Insurance, Nutrient, Satellite, Sonar, Wine

Department of Electronics & Communication Engineering

Course Overview & Instructions

This assignment spans all six units of the Data Analytics course curriculum. Students are required to complete each question using the specified datasets and visualization tools. Document your observations, code snippets, and visual outputs in a single lab report.

Datasets Provided

Dataset	Rows	Cols	Description
HR_comma_sep.csv	14,995	10	Employee HR metrics – satisfaction, evaluation, salary, attrition etc.
insurance.csv	1,338	7	Medical insurance charges with age, BMI, smoker status, region.
nutrient.csv	27	6	Food nutrition values – energy, protein, fat, calcium, iron.
Satellite.csv	6,435	37	Multi-spectral satellite pixel values for land-use classification.
Sonar.csv	208	61	Sonar signal features (60 frequency bands) for rock/mine classification.
wine.csv	178	14	Chemical properties of three classes of Italian wine.

General Instructions

- All code must be executed in Google Colab unless stated otherwise. Attach the Colab notebook link in your report.
- Visualizations created in Tableau / Power BI should be exported and included as screenshots.
- Label all graphs with proper titles, axis labels, and legends.
- Submit a single PDF report along with the source notebook (.ipynb).
- Academic honesty is expected; plagiarism will result in zero marks.

Question 1 — Understanding Data & Attribute Classification

Dataset	insurance.csv
Unit Coverage	Unit I – Understanding Data & Data Processing
Tools	Google Colab (Python · Pandas · Matplotlib)
Marks	[15 Marks]

Background

The insurance dataset contains 1,338 records describing patient demographics and their medical insurance charges. The features include age, sex, BMI, number of children, smoker status, region, and charges. This dataset provides an excellent opportunity to understand different attribute types and perform foundational data processing tasks.

Tasks

Part A – Attribute Type Identification (3 Marks)

Examine the insurance dataset and classify each of its seven attributes into one of the following types:

- Nominal – categories with no inherent order
- Ordinal – categories with a meaningful order
- Discrete – countable numeric values
- Continuous – measurable numeric values on a continuous scale

Present your classification in a table and justify each decision with a one-line explanation.

Part B – Data Extraction & Cleaning (4 Marks)

1. Load the dataset in Google Colab using Pandas. Display the first 10 rows and the dataset info (df.info()).
2. Check for and report any missing values or duplicate records. If present, handle them appropriately and justify your approach.
3. Identify and document any obvious data inconsistencies (e.g., negative charges, unrealistic BMI values).
4. Convert the categorical attribute smoker into a binary numeric encoding (0/1) and region into numeric codes using pandas.get_dummies(). Explain the transformation.

Part C – Data Viewing (4 Marks)

Using the cleaned dataset, perform the following:

5. Geometric View: Plot a 2D scatter plot of BMI (x-axis) vs. charges (y-axis). Color-code data points by smoker status (yes/no). Comment on the pattern you observe.
6. Algebraic View: Construct the data matrix for the first 10 rows using only the numeric attributes (age, bmi, children, charges). Display it as a formatted table.
7. Probabilistic View: Plot the probability distribution (histogram with KDE curve) of the charges column. Identify whether the distribution is skewed. Use Matplotlib/Seaborn in Colab.

Part D – Correlation Analysis (4 Marks)

Compute and visualize the correlation matrix for all numeric attributes in the insurance dataset.

8. Use df.corr() to generate Pearson correlation coefficients.
9. Display a heatmap using Seaborn with annotations.
10. Identify the pair of attributes with the highest positive correlation and the pair with the highest negative correlation.
11. Discuss, in two paragraphs, what these correlations imply about the relationships between patient demographics and insurance charges.

Expected Deliverables

- Attribute classification table (Part A).
- Python code for cleaning and transformations (Part B).
- Three visualizations: scatter plot, data matrix display, histogram with KDE (Part C).
- Correlation heatmap with written analysis (Part D).

Question 2 — Data Objects, Attribute Types & Statistical Descriptions

Dataset	HR_comma_sep.csv
Unit Coverage	Unit II – Data Objects & Attribute Types | Unit III – Basic Statistical Descriptions
Tools	Tableau / Power BI + Google Colab
Marks	[20 Marks]

Background

The HR dataset contains 14,995 records of employee profiles, capturing metrics such as satisfaction level, last evaluation score, number of projects, average monthly hours, tenure, work accidents, whether the employee left the company, promotions in the last five years, department, and salary. This dataset is rich with quantitative, categorical, and binary attributes, making it ideal for statistical exploration and visualization.

Tasks

Part A – Data Object Classification (4 Marks)

Classify the HR dataset using the data type taxonomy from Unit II. For each relevant category below, identify which columns belong to that type and give a brief justification:

- Quantitative multidimensional data (e.g., continuous numeric features)
- Categorical and mixed attribute data
- Binary and set data
- Non-dependency-oriented vs. dependency-oriented data (explain whether employee records could exhibit temporal or network dependencies)

Part B – Central Tendency & Dispersion (6 Marks)

Using Google Colab (Pandas and NumPy), compute the following for the columns satisfaction_level, last_evaluation, number_project, average_montly_hours, and time_spend_company:

12. Mean, Median, and Mode (Measuring Central Tendency)
13. Range, Variance, Standard Deviation, and Interquartile Range (Measuring Dispersion)
14. Present results in a clean summary table. Comment on which columns are highly dispersed versus tightly clustered.

Part C – Univariate & Multivariate Analysis (5 Marks)

15. Univariate Analysis: For the satisfaction_level column, plot a quantile plot (Q-Q plot), histogram, and box plot in Google Colab. Identify and discuss any outliers.
16. Multivariate Analysis: Create a pair plot (using Seaborn pairplot) for the numeric columns, color-coded by the left (attrition) column. Identify two pairs of attributes that show visible separation between retained and departed employees.
17. Compute and display the correlation coefficient between last_evaluation and number_project. Interpret the result.

Part D – Tableau / Power BI Dashboard (5 Marks)

Import the HR dataset into Tableau Public or Power BI Desktop and build a dashboard containing:

- A bar chart of the count of employees who left vs. stayed, grouped by Department.
- A histogram of satisfaction_level distribution separated by salary band (low / medium / high).
- A scatter plot of last_evaluation vs. average_montly_hours, color-coded by left.

Export the dashboard as a PNG/PDF and include it in your report. Write a paragraph summarizing the key HR insights visible in the dashboard.

Expected Deliverables

- Data object classification table (Part A).
- Statistical summary table with mean, median, mode, variance, SD, IQR (Part B).
- Q-Q plot, histogram, and box plot for satisfaction_level; pair plot (Part C).
- Tableau / Power BI dashboard screenshot + written analysis (Part D).

Question 3 — Measuring Data Similarity & Dissimilarity

Datasets	wine.csv + nutrient.csv
Unit Coverage	Unit IV – Measuring Data Similarity and Dissimilarity
Tools	Google Colab (Python · SciPy · Sklearn) + Intelligent Cloud / AWS
Marks	[20 Marks]

Background

Understanding how similar or dissimilar data objects are is fundamental to clustering, classification, and anomaly detection. In this question you will compute proximity measures on two datasets with contrasting attribute profiles: the wine dataset (all continuous chemical measurements) and the nutrient dataset (a small mixed dataset with a nominal food name and continuous nutritional values).

Tasks

Part A – Data Matrix vs. Dissimilarity Matrix on Wine Data (6 Marks)

18. Load wine.csv in Google Colab. Select the first 10 rows and six numeric attributes: Alcohol, Malic, Ash, Magnesium, Phenols, Flavanoids.
19. Display the 10 × 6 data matrix as a formatted DataFrame.
20. Compute and display the 10 × 10 Euclidean dissimilarity matrix using scipy.spatial.distance.cdist. Render it as a heatmap using Seaborn.
21. Repeat using Manhattan (L1) distance. Compare visually. Which distance metric produces more extreme dissimilarity values? Why?

Part B – Minkowski Distance Experiment (5 Marks)

Choose any three wine samples (rows) and compute pairwise Minkowski distance for p = 1, 2, 3, and 10.

22. Tabulate the results for all three pairs across the four values of p.
23. Plot a line chart: x-axis = p value, y-axis = Minkowski distance, with one line per sample pair.
24. Explain the effect of increasing p on the distance value. What happens as $p \to \infty$? (Chebyshev distance)

Part C – Proximity for Nominal & Binary Attributes on Nutrient Data (5 Marks)

The nutrient dataset contains the nominal attribute Food_Item alongside continuous nutritional measurements.

25. Proximity for Nominal Attributes: Using simple matching, compute the similarity between any two food items based only on whether their energy group (categorize energy into Low / Medium / High) and their fat group (Low / Medium / High) match. Show the calculation step by step.
26. Binary Attributes: Create a binary attribute high_protein (1 if protein > median, else 0) and high_iron (1 if iron > median, else 0) for all food items. Compute the Jaccard similarity and Simple Matching Coefficient (SMC) between three pairs of food items. Discuss the difference between Jaccard and SMC in the context of asymmetric binary attributes.

Part D – Cloud Deployment (4 Marks)

Upload the wine dataset to AWS S3 (free tier) or use Google Colab with Google Drive. Implement the Minkowski distance computation as a reusable Python function and host the notebook in the cloud environment. Capture a screenshot of the running notebook in the cloud environment and include it in your report.

Expected Deliverables

- Data matrix and dissimilarity heatmaps for Euclidean and Manhattan (Part A).
- Minkowski distance table and line chart (Part B).
- Nominal proximity and Jaccard/SMC calculations with discussion (Part C).
- Cloud-hosted notebook screenshot (Part D).

Question 4 — Data Pre-processing Pipeline

Dataset	Sonar.csv + HR_comma_sep.csv
Unit Coverage	Unit V – Data Pre-processing
Tools	Google Colab (Sklearn · Pandas) + Power BI for visualization
Marks	[20 Marks]

Background

Raw data is almost never ready for direct analysis. The Sonar dataset contains 208 instances with 60 continuous frequency-band features and one binary class label (R = Rock, M = Mine). Its high dimensionality makes it an ideal candidate for dimensionality reduction. The HR dataset, with its mix of numeric and categorical features, exercises data integration and transformation techniques.

Tasks

Part A – Data Cleaning (4 Marks)

Using HR_comma_sep.csv:

27. Check for missing values, duplicate rows, and outliers in the numeric columns (use IQR-based outlier detection). Report counts.
28. Remove duplicates. For outliers in average_montly_hours, apply winsorization (capping at the 5th and 95th percentile). Compare histograms before and after.
29. Check data type consistency. If any column has an unexpected data type (e.g., numeric stored as object), fix it and document the change.

Part B – Data Integration & Transformation (4 Marks)

Working with HR_comma_sep.csv:

30. Simulate data integration by splitting the dataset into two halves (first 5 columns; last 5 columns) and re-merging them on a common index. Verify no data loss.
31. Apply min-max normalization to satisfaction_level, last_evaluation, and average_montly_hours. Display the before/after statistics.
32. Apply z-score standardization to the same columns. Plot side-by-side box plots (original, min-max normalized, z-score standardized) for each column.
33. Perform equal-width data discretization on the continuous column last_evaluation into 4 bins. Label the bins Unsatisfactory, Average, Good, Excellent. Show the value counts per bin.

Part C – Dimensionality Reduction on Sonar Data (7 Marks)

34. Load Sonar.csv. Separate features (V1–V60) from the class label. Apply StandardScaler to normalize the features.
35. Apply PCA (Principal Component Analysis) using sklearn.decomposition.PCA. Plot the explained variance ratio for each component as a bar chart. How many components explain 90% of the total variance?
36. Reduce the dataset to the optimal number of components identified above. Print the shape of the reduced dataset.

37. Plot a 2D scatter plot of the first two principal components, color-coded by the Rock/Mine class label. Comment on whether the two classes are separable in the 2D PCA space.
38. Compare PCA with t-SNE (sklearn.manifold.TSNE with n_components=2). Plot both 2D embeddings side by side and discuss which better separates the classes.

Part D – Feature Extraction & Reporting in Power BI (5 Marks)

After pre-processing, export the cleaned and PCA-reduced Sonar data as a CSV. Import it into Power BI and create:

- A table visual showing the top 5 PCA component loadings (feature importance).
- A 100% stacked bar chart showing the proportion of Rock vs. Mine samples before and after PCA reduction (to confirm class balance is maintained).

Include screenshots in your report with a short paragraph of findings.

Expected Deliverables

- Before/after histograms for outlier removal, type-fix documentation (Part A).
- Normalization/standardization statistics and side-by-side box plots (Part B).
- PCA explained variance chart, 2D PCA scatter plot, t-SNE comparison (Part C).
- Power BI visuals screenshots + analysis paragraph (Part D).

Question 5 — Exploratory Data Analysis & Pattern Mining

Dataset	Satellite.csv
Unit Coverage	Unit VI – Exploratory Data Analysis
Tools	Google Colab (Sklearn · Matplotlib) + Tableau for dashboard
Marks	[15 Marks]

Background

The Satellite dataset contains 6,435 instances, each representing a 3×3 pixel neighbourhood in a multi-spectral satellite image with 36 spectral features. The target attribute classes encodes land-use type (1 = Red Soil, 2 = Cotton Crop, 3 = Grey Soil, 4 = Damp Grey Soil, 5 = Soil with Vegetation Stubble, 7 = Very Damp Grey Soil). Performing EDA on remote-sensing data is directly relevant to ECE students working on image processing and embedded sensing systems.

Tasks

Part A – Data Loading & Initial EDA (3 Marks)

39. Load Satellite.csv (note: the file uses semicolons as delimiters). Check and report the shape, column names, data types, and class distribution (value counts of the classes column).
40. Plot a pie chart and a bar chart of the class distribution. Are the classes balanced?
41. Check for any missing values and report findings.

Part B – Clustering (5 Marks)

Apply K-Means clustering (ignore the true class labels for this part):

42. Scale the 36 spectral features using StandardScaler.
43. Use the Elbow Method (plot inertia for k = 2 to 10) to determine the optimal number of clusters. Identify the elbow point.
44. Fit K-Means with the optimal k. Plot a 2D scatter of the first two features colored by cluster assignment.
45. Compute and report the Silhouette Score for your chosen k. Compare it with k = 3 and k = 7.
46. Cross-tabulate the K-Means cluster labels against the true class labels. Do the clusters align with the true classes? Discuss.

Part C – Classification & Outlier Detection (4 Marks)

47. Split the dataset into 80/20 train-test sets (stratified). Train a Decision Tree Classifier (max_depth=5) and report accuracy, precision, recall, and F1 score (per class and macro-averaged).
48. Plot the confusion matrix as a heatmap.
49. Apply Isolation Forest (sklearn.ensemble.IsolationForest, contamination=0.05) to detect outliers in the spectral features. Report the number of anomalies detected. Plot those anomalous points in a scatter plot of feature x1 vs. x2, highlighted in red.

Part D – Tableau Dashboard (3 Marks)

Using the Satellite dataset in Tableau Public, create a dashboard containing:

- A heat map of average x1 and x2 feature values by land-use class.
- A stacked bar chart comparing true class distribution vs. K-Means cluster distribution.

Export and include in your report. Write a brief paragraph on what the EDA reveals about spectral separability of land-use types.

Expected Deliverables

- Class distribution charts (Part A).
- Elbow plot, K-Means scatter, silhouette scores, cross-tabulation (Part B).
- Classification report, confusion matrix heatmap, isolation forest scatter (Part C).
- Tableau dashboard screenshot + written summary (Part D).

Question 6 — Integrative Mini-Project: End-to-End Analytics Pipeline

Datasets	Choose any TWO from: insurance.csv, wine.csv, HR_comma_sep.csv
Unit Coverage	All Units I–VI (Integrative)
Tools	Google Colab + Tableau OR Power BI + AWS S3 or Intelligent Cloud (at least two tools required)
Marks	[10 Marks — Bonus / Viva Component]

Background

This question synthesizes all six units of the course into a coherent end-to-end analytics pipeline. You will select two datasets, define a data-driven problem statement, pre-process the data, perform exploratory analysis, apply at least one machine learning technique, and present findings using cloud-hosted visualizations. This mirrors real-world data analytics workflows in industry.

Tasks

Step 1 – Problem Formulation (1 Mark)

Write a concise problem statement (4–6 sentences) describing:

- The domain context of your chosen datasets.
- The analytical question you intend to answer (e.g., 'Can we predict whether an employee will leave based on HR metrics?').
- The evaluation criteria you will use to measure success.

Step 2 – Data Understanding & Pre-processing (3 Marks)

50. Combine or use both datasets (integrate if applicable). Document the data types and attribute classifications for each (Unit I, II).
51. Perform data cleaning: handle missing values, outliers, and type issues (Unit V).
52. Apply appropriate transformations: normalization, encoding, discretization (Unit V).
53. Generate a data quality report (a markdown table in Colab listing: column, original dtype, action taken, final dtype).

Step 3 – Exploratory & Statistical Analysis (3 Marks)

54. Compute all central tendency and dispersion measures for key numeric columns (Unit III).
55. Plot at least three of: histogram, box plot, scatter plot, Q-Q plot, correlation heatmap (Unit III, VI).
56. Compute at least one similarity / dissimilarity measure between data objects (Unit IV).

Step 4 – Modelling & Insights (2 Marks)

57. Apply one algorithm: K-Means clustering, Decision Tree classifier, or outlier detection (Unit VI).
58. Report performance metrics relevant to your chosen algorithm.
59. Summarize findings in a half-page written narrative: what did the data reveal? Were your expectations confirmed?

Step 5 – Cloud Dashboard (1 Mark)

Publish your Colab notebook to GitHub or Google Drive (shared link) and build a final dashboard in Tableau Public or Power BI Service (cloud version). Submit the public link. The dashboard must be accessible without login.

Evaluation Rubric for Assignments

Criterion	Full Marks	Partial Credit
Clear problem statement with domain context	**1 Mark**	0.5 – vague or missing context
Thorough pre-processing with documented steps	**3 Marks**	1–2 – some steps skipped/undocumented
Correct statistical analysis & visualizations	**3 Marks**	1–2 – errors or missing plots
ML model applied with valid evaluation	**2 Marks**	1 – model run but no metrics
Cloud-hosted, accessible dashboard/notebook	**1 Mark**	0 – local only

Appendix – Visualization Tool Quick Reference

The following table summarises the primary use cases for each tool in the context of this assignment:

Tool	Best For	Questions Used In
Google Colab	Python-based data cleaning, statistics, ML, PCA, t-SNE, clustering	Q1, Q2, Q3, Q4, Q5, Q6
Tableau Public	Interactive dashboards, bar charts, scatter plots, heatmaps for any dataset	Q2, Q5, Q6
Power BI	Business-style dashboards, HR analytics, drilldown reports	Q2, Q4, Q6
AWS S3	Cloud storage for datasets; notebook hosting; scalable processing	Q3, Q6
Intelligent Cloud	Azure ML / Google Cloud AI for scalable modelling and pipeline hosting	Q3, Q6
Flunk / Others	Supplementary tools as required by instructor	As directed

Recommended Python Libraries

- pandas, numpy – data loading, cleaning, statistics
- matplotlib, seaborn – all visualizations in Colab
- scikit-learn – pre-processing, PCA, K-Means, Decision Tree, Isolation Forest, t-SNE
- scipy – distance/dissimilarity metrics
- plotly – interactive charts (optional but encouraged)